

CTAB DNA extraction

Description

Extracting DNA is a very secure work; contamination is easy and will result in untrustworthy data. DNA-extraction takes a lot of time, work secure so you won't waste two days of work. Some chemicals are very hazardous, so work in the flow chamber as much as possible, always wear gloves (also avoids contamination). Timing is important in some steps, possible stops are indicated, don't stop in other phases of the protocol.

Product Components

- 2% CTAB isolation buffer (pH 8)
- 20 mg/ml proteinase K
- 1 mg/ml RNase
- chloroform
- iso-amyl-alcohol (IAA)
- 5 M NaCl
- ice cold 96% ethanol for precipitation
- ice cold 80% ethanol for washing
- sterile milliQ water
- 0.5 ml tubes
- 56°C water bath or thermal cycler
- sterile grinders for 0.5 ml tubes for mites and insects only

Protocol

A. Preparation of solutions

1. Defrost proteinase K and 1 mg/ml RNase
2. Prepare enough fresh chloroform: IAA = 24:1 extraction solution for all your samples.

B. DNA isolation from animal tissue

Before DNA isolation remove most of the ethanol or acetone and vacuum dry the samples 10-minutes

1. Prepare 1.5ml tubes with ~~200~~ 2%CTAB buffer with ~~2.5~~ proteinase-K. *→ 200 200 µl 200 µl 600 µl 200 µl*
2. Place your tubes on RT and add a small piece of thorax or leg (1-3) in the tube.
3. Crush the tissue with a clean pestle, twist and turn the pestle till the tissue is gone. Note: add extra 2%CTAB buffer if a lot of tissue is used.
4. Vortex short and incubate 1 to 2 Hour at 55°C. Shake (gently!)
5. Add equal volume chloroform: IAA (24:1) and vortex 2 times 5 seconds. *→ 800 µl*
6. Separate phases by spinning for 10 min at 12000 rpm.
7. Transfer supernatant (~~450 µl~~) to a fresh sterile tube *→ 600 µl*
8. and add ~~250 µl~~ (~~275 µl~~) ~~ice cold 100% EtOH~~ *→ 100 µl / 1X isopropanol 2-methanol (600 µl)*
9. Shake and incubate on ice for approx. 5 minutes
10. Centrifuge for 10 min. at 13000rpm (check position of tubes)
11. Carefully remove supernatant with pipet
12. Dissolve the pellets in 100 µl st.H₂O, add ~~1 µl~~ 5 µl RNAse (1 mg/ml!) and 2.5 µl *2 µl Tris*
5M NaCl. Incubate at 37°C for at least 0.5 hours (RNAse plus incubation time are optional. Possible stop).
13. Add 250 µl of 96% ethanol. Mix gently. Place the samples on ice for 5 minutes (possible stop)
14. Spin for 10 minutes at 13000 rpm. Carefully remove supernatant with pipet
15. Wash with 80% ethanol; add 500 µl ethanol, swirl gently, spin for 5 minutes at max speed (13000), Carefully remove supernatant with pipet
16. Dry pellet in a speedvac for 10 minutes
17. Dissolve the DNA in 10 to 50 µl st.H₂O (for higher concentrations take less volume ~~20~~ 20 µl). Incubation in the 37°C waterbath will speed up the solution of the DNA. *→ 20 µl*
18. Check 2 µl DNA on a 1.5% agarose gel.

C. Cleaning.

1. Clean the centrifuge with a moisturized paper towel.
2. Clean your workspace with moisturized paper towel and a bit of 76% cleaning ethanol.

References

Doyle, Jeff (1991). Molecular Techniques in Taxonomy.